

BIO3SA3: Applied Statistics for Biology

UNIT 1: Introduction

Outline

Course structure

Prof

TA

Students

Philosophy

Communication

- ▶ Check Avenue frequently for announcements and new information
 - ▶ I will aim to share information promptly and clearly on Avenue
- ▶ The professor is Jonathan Dushoff
 - ▶ Contact via forum on Avenue
 - ▶ What do you think?

Expectations of professor

- ▶ Start and end on time
- ▶ Focus on understanding and applications, not memorization
- ▶ Open to questions – in class (when possible); office hours; forum
 - ▶ What is a good structure for office hours?

Expectations of students

- ▶ Start and end on time
- ▶ Don't talk while other students are talking, or while I am responding to student questions
- ▶ If you must talk at other times, be unobtrusive
- ▶ Don't use the internet for non-class activities

Taking notes

- ▶ You will do best if you take notes
 - ▶ You should know by now what works for you
 - ▶ Or else that you need to keep working on it
- ▶ If a new concept is making sense to you right now, write something that will help you remember
- ▶ If there's something specific I think you all need to write down, I will write it for you (or mark it as an answer)

What you need to know

- ▶ I am still designing this course, based on what you-all know and what you want to learn
- ▶ My basic goals are
 - ▶ Build computer and data-management skills
 - ▶ Convince you that you need a statistical philosophy to do science
 - ▶ Learn basic principles of this philosophy
 - ▶ Discuss principles of different tests
 - ▶ linear models, generalized linear models, permutation tests
 - ▶ random effects and mixed models
 - ▶ maybe other things

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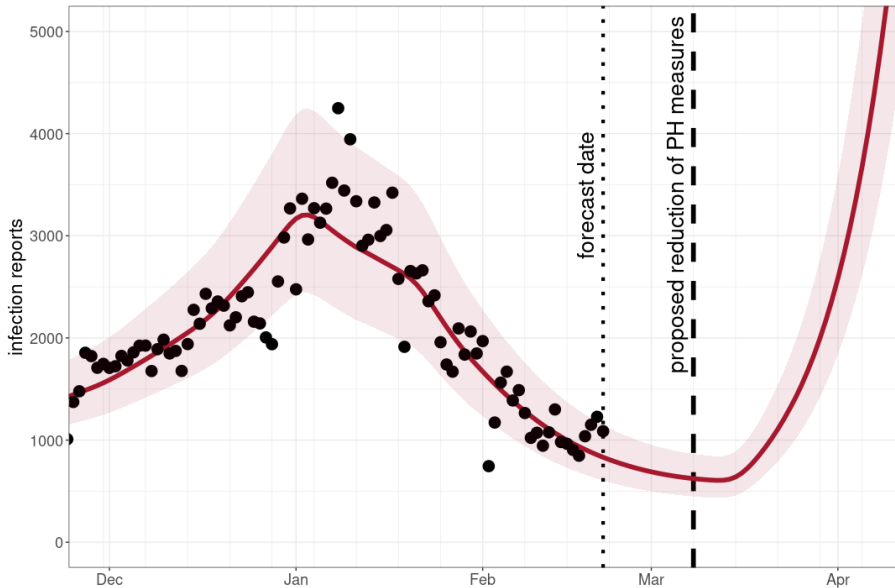
TA

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Philosophy

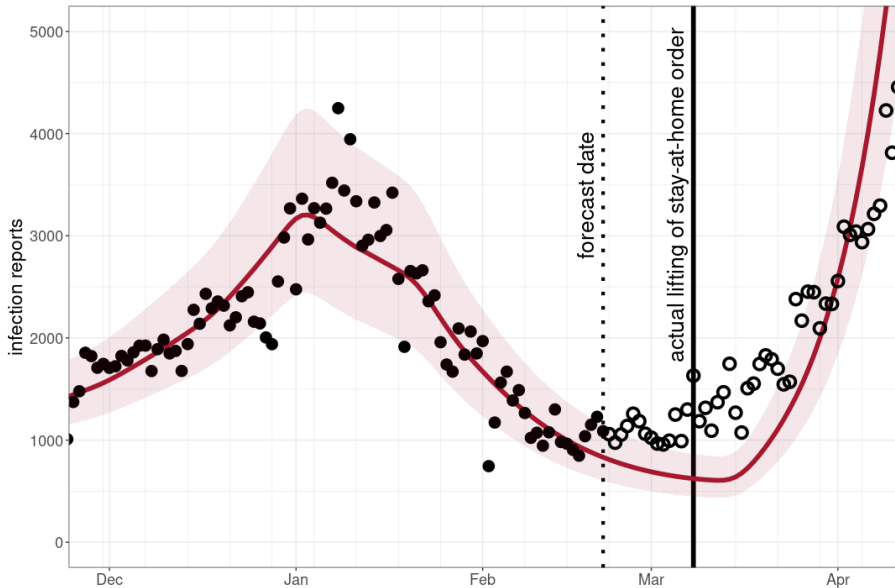
Disease research (present)

McMasterPandemic forecast from 21 Feb 2021

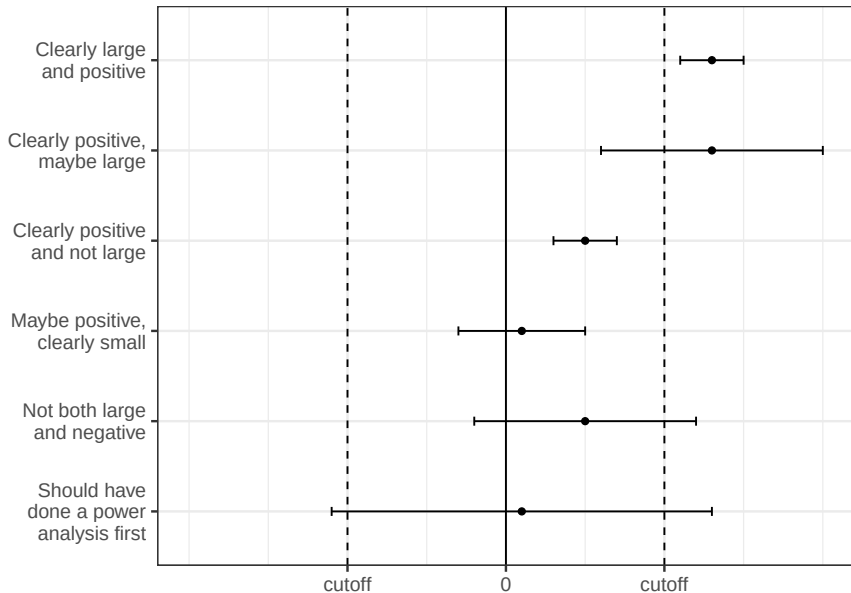


Disease research (present)

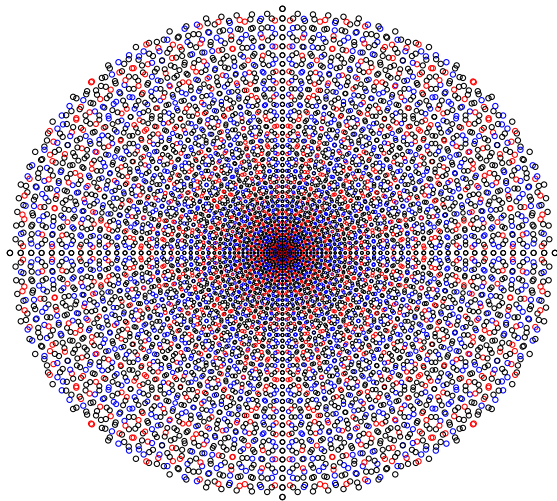
McMasterPandemic forecast from 21 Feb 2021



Statistical philosophy (present)



Recreational math (present)



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Students

- ▶ A broad range of background, interest and experience
- ▶ Not everyone will be happy all the time
- ▶ Not everyone did the quiz

Learning goals

- ▶ Statistical philosophy; where stats meets science
- ▶ Practical computer skills
- ▶ Practical stats skills (but how far can we get?)

Questions

► * Winner

Participation

- ▶ Actually talking
- ▶ Index cards
- ▶ Peerwise

Midterm timing

- ▶ Tue 20 Oct and 24 Nov

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What is the definition of a P value?

Why is this a good definition?

I measure plant growth in plots where I did and did not add K

- ▶ What are some concerns before I even analyze the data?
- ▶ What does it mean if $P < 0.05$?
- ▶ What does it mean if $P > 0.05$?

Significance

- ▶ Is statistical significance a statement about significance (importance) or something else?
- ▶ Is statistical significance a statement about the study system, or about the data set?

Statistics is not magic!

- ▶ Scientists are sometimes intimidated
- ▶ Scientists sometimes *want* statistics to be magic

Index card

- ▶ Write your macid (mostly letters) on the blank side
- ▶ Write one brief question and one brief comment on the lined side
 - ▶ Neatly, if you can